

SMART ISSUE TRACKER

Quality Assurance & Verification Report

Project Name:	Smart Issue Tracker	Release Version:	1.0.0 (Production-Ready)
Testing Scope:	Full System Verification	QA Lead Engineer:	Vasiharan
Date of Testing:	June 10, 2026	Status:	APPROVED FOR RELEASE

1. Testing Scope & Methodology

This verification report covers the testing of the **Smart Issue Tracker** system, focusing on its hybrid duplicate detection pipeline. The pipeline uses localized **FAISS (Facebook AI Similarity Search)** cosine similarity checks on 384-dimensional dense vectors (generated via all-MiniLM-L6-v2) combined with **Google Gemini 1.5/2.0 Flash** Generative AI model verification. Testing was carried out across functional interfaces (React Customer Intake, Admin Dashboard, Categories CRUD, History logs), backend APIs, security layers (session authentication), data constraints, invalid inputs, and fallback procedures.

2. Test Environment Details

Component	Technologies & Configurations
Operating System	Windows 10/11 (Local Workstation)
Frontend Framework	React 18, TypeScript, Vite, Tailwind CSS
Backend Framework	Django 5.0, Django REST Framework (DRF)
Vector Database	FAISS (CPU-only index: Flat Inner Product)
Embedding Model	Sentence-Transformers (all-MiniLM-L6-v2, 384-d)
Generative Model	Google Generative AI SDK (gemini-1.5-flash)
Primary Database	PostgreSQL 14 / SQLite3

3. Executive Test Execution Summary

Total Test Cases	Passed	Failed	Pass Rate
15	15	0	100.0%

4. Test Coverage & Module Results

Module Name	Test Scenario / Goal	Status
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Ticket Intake	Verify ticket submissions and generation of UNIQUE tickets.	PASSED
Deduplication	Verify auto-duplicate merges on highly similar tickets ($\geq 90\%$).	PASSED
Gemini AI Core	Verify semantic verifications and structured JSON response maps.	PASSED
Fallback Pipeline	Verify local FAISS duplicates fallback when Gemini is offline.	PASSED
Admin Controls	Verify dynamic category CRUD and administrative overrides.	PASSED
Security/Auth	Verify auth checks and session blocking on secure resource urls.	PASSED
UX / Metrics	Verify dashboard chart aggregations and system audit logs.	PASSED

5. QA Assumptions & Rules

* **FAISS Thresholds:** Cosine similarity score $\geq 90\%$ merges duplicates immediately. Scores between 75% and 90% query Gemini AI. Scores $< 75\%$ are immediately marked UNIQUE. * **Gemini Quota / Connectivity:** If the Gemini API is rate-limited (429) or offline, the system safely falls back to FAISS vector scores as a secondary validation to prevent ticket ingestion blockages. * **Database Isolation:** Unit tests use an isolated database state to prevent interference with active FAISS indexing files.

6. Quality Assurance Conclusion

The **Smart Issue Tracker** system successfully passed all 15 functional, logical, integration, and security test cases. The hybrid vector-LLM deduplication pipeline demonstrated high stability, failing over gracefully under simulated API blocks, and updating FAISS vector representations and DB indexes synchronously. The product is verified as robust and is **fully approved for deployment and final portfolio submission.**